

2. Machine learning (ML)/artificial intelligence (AI)

The next job that is in high demand in 2019 is for ML/AI professionals. According to the article in the *Economic Times*, there are 50,000+ job vacancies for ML/AI related jobs. Shortage of skilled resources in these fields has led to very many vacancies. Machine learning and AI require people to have programming and mathematical skills which could be used to train systems for decision making. The skills needed are programming skills (Python, Java, C++, R, etc), a strong foundation in mathematics covering probability and statistics, strong algorithm building skills and above all, curiosity, patience and perseverance. The median salary for ML/AI job roles in India is ₹ 1.43 million (14.3 lakhs) per annum.

3. Cloud architects/developers

There is no doubt that every industry is aiming to have its infrastructure and applications hosted in the cloud. This brings us to the next in-demand job role – the cloud architect/developer. The architect helps an organisation to design a possible infrastructure hosted in the cloud while the developer creates the applications that run on it. India will have one million cloud professionals by 2022. While there are several cloud providers who have their own knowledge repositories, some of the most basic skills such as virtualisation, containerisation, serverless computing, NoSQL DB and programming skills are necessary for every aspiring cloud architect/developer. The entry-level salary for a cloud professional could be in the range of ₹ 1.2-1.9 million (12-19 lakhs) per annum.

4. Cyber security experts

As technology is evolving, so are the modes of attack employed by malicious applications. Thus the challenge is to provide a secure means for the current applications to operate, bearing in mind that the Internet is of prime importance

to all organisations. It is estimated that the Indian cyber security market will be around US\$ 35 billion by 2025 and at present, more than 30,000 jobs are vacant across the country. The skills required for the job are the ability to learn about various types of security threats, to be able to hack into systems, programming skills and the ability to adapt to the situation. The median salary for a cyber security professional is around ₹ 1.1-1.9 million (11-19 lakhs) per annum while top-notch experts in the field can draw ₹ 2-4 million per annum.

5. IoT engineer

We are living in an age when almost all devices are being connected to the Internet, which is why the IoT engineers enter the picture. They design, develop and program the devices so that the latter can interact with the Internet. The demand for IoT engineers has been rising consistently. The period from 2014 to 2016 saw a 300 per cent rise in demand for IoT professionals. The global IoT market is expected to grow to US\$ 300 billion by 2025, as per the latest statistics. The Indian government's promise to invest US\$ 70 billion in smart cities makes this job role even more in demand. The skills that are required vary from programming to integrating small devices such as Raspberry Pi, Arduino, etc, to various other systems; or building a new smart system from scratch that could be connected to the Internet. In India, the median salary of an IoT professional, according to November 2019 statistics, is ₹ 21,238 per month.

6. Digital transformation consultant

There are a great many things which are not yet digital. So a digital transformation engineer is required in order to make these digital. Most people and businesses are yet to become digitally connected and this is what a digital transformation consultant's role is all about. Many business processes are still conducted in an old fashioned

manner and hence are on the verge of losing out to global competition. The demand for digital transformation consultants is rising and such professionals come with a variety of skills covering IoT, Web development, Big Data, social media platform engagement, etc. The median salary for a digital transformation consultant is ₹ 850,000 (8.5 lakhs) per annum.

7. Program manager

With the digital revolution going on across the world, the demand for program managers has also increased. It is quite impossible for any organisation to roll out a new policy or a program without naming a program manager who will take charge – to carry out the policy and the entire program. Most organisations that are dealing with challenges such as digital transformation, integrating ML/AI, etc, require a program manager who will oversee such work and evaluate all the possible opportunities and challenges. This job role requires experience in handling four to five projects and staying updated with all that's new in software development as well as in digital technology. The salaries can vary depending on the experience and the organisations; however, the entry-level program manager can expect salaries in the range of ₹ 2.9-3.6 million (29-36 lakhs) per annum.

8. AR/VR developer

The next tech revolution to unfold is augmented reality and virtual reality. It is a new field which has seen growth in the demand for developers. While augmented reality uses image capture to get various information from the image, virtual reality deals with tools and gadgets that simulate a different world. Both have seen encouraging usage in several fields, especially the medical domain, where doctors can simulate an emergency operation

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